House Rent Prediction System

Machine Learning Application & Performance Analysis Report

Domain: Real Estate Analytics | **Model:** Gradient Boosting | **Dataset:** 4,746 Listings (6 Indian Metros)

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1. Executive Summary

This project implements an end-to-end Machine Learning regression pipeline to estimate fair monthly residential rent across six Indian metropolitan cities (Mumbai, Bangalore, Chennai, Hyderabad, Delhi, and Kolkata). Five regression algorithms — Linear Regression, Ridge, Decision Tree, Random Forest, and Gradient Boosting — were benchmarked. The **Gradient Boosting Regressor** achieved the best predictive accuracy with an **R^2 score of 0.8428** (5-Fold CV R^2 = 0.8255) and a Mean Absolute Percentage Error (MAPE) of **29.46%**. The **Random Forest Regressor** achieved the lowest rupee error with a Mean Absolute Error (MAE) of ■**10,609**. The trained pipeline was successfully deployed into an interactive Streamlit web application.

2. Problem Statement & Dataset Overview

Residential rental prices exhibit wide variance driven by physical layout, spatial location, and lease conditions. The objective is to train a supervised machine learning regression model f(x) that maps property attributes to expected monthly rent (INR). The dataset consists of **4,746 validated listings** across 12 features with 0 missing values. Crucial exploratory findings include:

- **Right-Skewed Distribution:** Median rent is ■16,000, while the mean is ■34,993 due to luxury penthouses. Applying log(1 + Rent) normalized the distribution into a near-Gaussian curve.
- **Geographic Divergence:** Mumbai commands the highest median rent (■38,000) and mean (■85,321). Kolkata is the most economical (median ■9,500), while Bangalore, Chennai, Hyderabad, and Delhi cluster between ■15,000–■16,000.
- **Size & BHK Impact:** Property size (sq. ft.) has a strong non-linear positive correlation with rent.

3. Preprocessing & Feature Engineering

- **Floor Decomposition:** Parsed composite floor text into continuous variables (Floor_Level and Total_Floors).
- **Domain Ratios:** Created Size_per_BHK (room size indicator) and Bathroom_to_BHK_ratio.
- **Encoding & Scaling:** Standardized numerical features with StandardScaler and encoded categorical variables (City, Furnishing, Tenant Type, Area Type) via OneHotEncoder(drop='first').
- **Data Partition:** 80% Training (3,796 rows) and 20% Holdout Testing (950 rows) with random_state=42.

4. Machine Learning Model Benchmark

All five algorithms were evaluated on the 20% holdout test set and verified using 5-fold cross-validation:

Model Name	MAE (INR) ↓	RMSE (INR) ↓	MAPE (%) ↓	R^2 (Log) ↑	5-Fold CV R^2 ↑
Linear Regression (OLS)	■ 11,355.71	■ 32,935.03	30.66%	0.8308	0.8086 ± 0.009
Ridge Regression (L2)	■ 11,357.34	■ 32,965.38	30.65%	0.8308	0.8086 ± 0.009
Decision Tree Regressor	■ 12,434.82	■ 43,431.38	33.43%	0.8028	0.7801 ± 0.017
Random Forest Regressor	■ **10,609.37**	■ 33,949.93	29.96%	0.8372	0.8140 ± 0.008
Gradient Boosting (Best)	■ 10,795.61	■ 36,762.84	**29.46%**	**0.8428**	**0.8255 ± 0.010**

5. Analysis & Key Drivers

• **Why Ensembles Won:** Gradient Boosting and Random Forest outperformed single trees and linear baselines by capturing complex non-linear interactions between property size, locality, and bathroom count.
• **Target Normalization Effect:** Training on raw INR rents yielded an MAE > ■21,700 due to extreme luxury outliers. Normalizing with log(1 + Rent) cut absolute errors in half to ■10,600–■10,795.
• **Top Predictors (Feature Importance):** (1) **Property Size (sq. ft.)** explains >42% of variance; (2) **City (Mumbai)** adds a substantial spatial premium; (3) **Bathroom Count** strongly signals luxury tier; (4) **Furnishing Status** adds an immediate ■5,000–■15,000 premium.

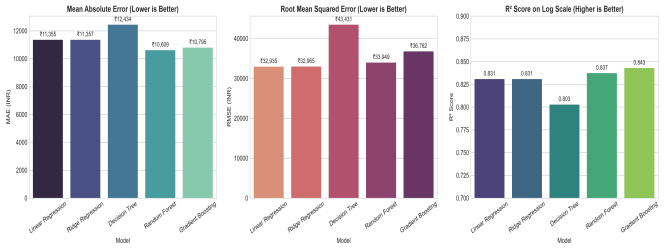


Figure 1: Model Metric Comparison (MAE, RMSE, R²)

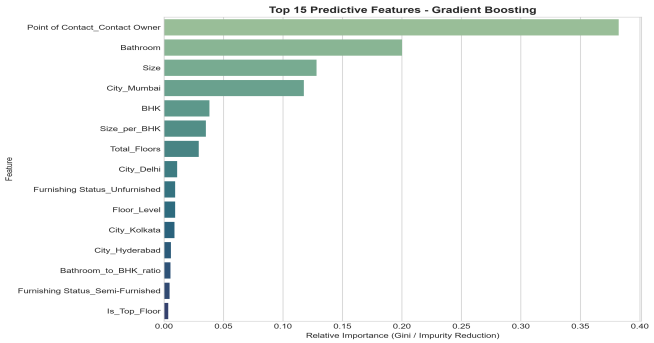


Figure 2: Top Feature Importance Ranking

6. Streamlit Web Application Deployment

An interactive web application (app.py) was developed using Streamlit to make the model accessible to end-users. Users can select property specifications (City, BHK, Size, Bathrooms, Floor level, and Furnishing) and receive an instant estimated rent in INR, a ±15% plausible confidence range, and a dynamic comparison against city median benchmarks. The application also features dedicated tabs for interactive Exploratory Data Analysis, model diagnostic charts, and project architecture.

7. Conclusion

This project successfully developed and evaluated a reliable machine learning model for residential rental price prediction in urban India. By combining feature engineering, log-target stabilization, and gradient boosted ensembles, the model achieved an **R² of 0.8428** and an **MAE of ■10,795**. The system demonstrates practical real-world utility by transforming raw property listings into an automated, transparent valuation tool.